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EU AI Act

European Union
Data Strategy:
Regulating AI-Data Flows

December 2025

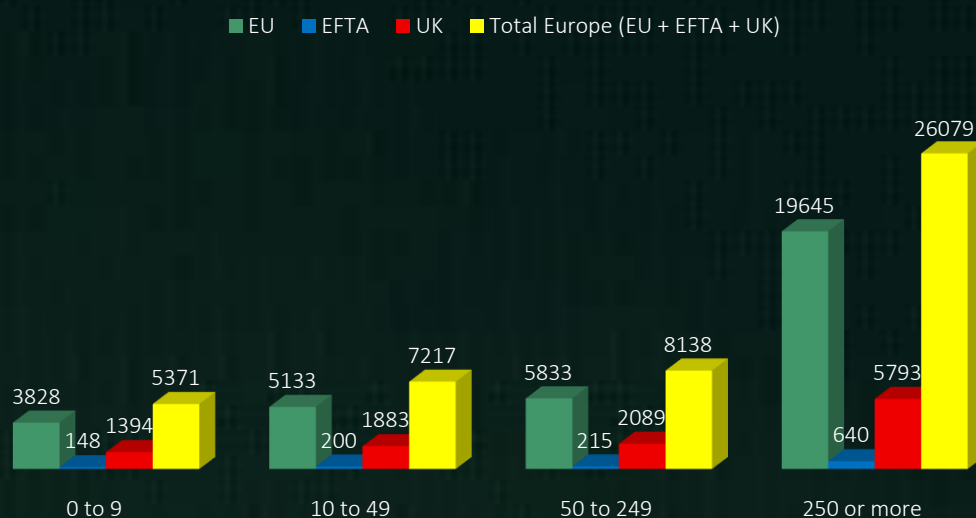
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Concentration of Enterprise Cloud Data Flows Among Large European Firms Highlights Urgent Need for Targeted AI Governance and GPAI Alignment Across SMEs

*European Commission: ECBDF volumes in 2023 by
enterprise size – EU, EFTA and UK (PB/Year) (MAR-2024)*



In 2023, Enterprise Cloud-Based Data Flows (ECBDF) across Europe reached 46,805 PB/year, with large enterprises (250+ employees) contributing 56% of total volumes. The EU accounts for the majority share, followed by the UK and EFTA. Small enterprises (0–9 employees), while numerous, generate relatively modest cloud traffic, highlighting a significant adoption gap in cloud-enabled operations.

This distribution underscores two critical implications. First, large enterprises are central to European cloud ecosystems and will be most impacted by regulatory developments under the EU AI Act, particularly regarding transparency, accountability, and risk management in high-volume data and AI applications. Smaller enterprises, though currently generating less cloud traffic, face an opportunity to scale responsibly, adopting GPAI-aligned practices early to avoid compliance bottlenecks.

Second, the uneven data flow distribution suggests targeted policy and infrastructure support could accelerate smaller enterprise adoption, fostering a more balanced digital economy. Constructively, enterprises should implement robust data governance frameworks aligned with the EU AI Act, ensuring model explainability, bias mitigation, and secure cross-border data flows. Integrating these practices into cloud operations will not only facilitate regulatory compliance but also strengthen competitive positioning in AI-driven markets.

Finally, aligning cloud strategy with GPAI model requirements—particularly in high-volume data environments—can transform ECBDF from a compliance challenge into a strategic asset, enabling scalable, responsible AI deployment across Europe’s enterprise landscape.



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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

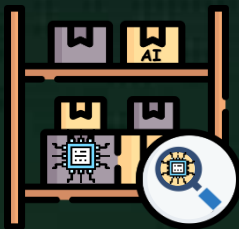
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contact: Team | contact@ai-and-partners.com

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Introduction

As the European Union Data Strategy takes shape, it represents a cornerstone initiative for building a unified, values-driven data economy across Europe. This comprehensive strategy emphasizes the creation of a single market for data—where data can be accessed, shared, and reused securely and fairly—while ensuring privacy, fostering innovation, and strengthening Europe’s digital sovereignty.

This report provides an in-depth guide for businesses, policymakers, and stakeholders navigating the opportunities and responsibilities under the European Data Strategy. It draws on best practices, global benchmarks, and actionable insights to support the successful implementation of the strategy across Member States. The report explores critical areas such as cross-sector governance, infrastructure investment, individual empowerment, and sectoral data spaces, aligning these efforts with the EU’s broader digital transformation agenda.

The European Data Strategy marks a decisive step toward creating an open, competitive, and trusted data ecosystem in Europe. By enabling secure and interoperable data flows across sectors and borders, it lays the groundwork for innovation in key industries, supports sustainable growth, and protects fundamental rights. This approach ensures that economic progress is balanced with ethical responsibility, reinforcing fairness, transparency, and accountability in the digital age.

Additionally, the report underscores the vital role of collaboration among public authorities, private enterprises, and civil society. Through shared governance, sectoral alliances, and investments in digital skills, stakeholders can unlock the full potential of data while safeguarding European values.

Whether you are an SME seeking new opportunities, a policymaker shaping data frameworks, or an enterprise driving digital innovation, this report offers a clear roadmap for aligning with the European Data Strategy and contributing to a robust, future-ready data economy.

Best regards,

Sean Musch

Founder/CEO

AI & Partners

A stylized, handwritten signature in white ink, appearing to read 'Sean Musch', is positioned to the right of the signature text. The background of the entire page features a dark green, abstract pattern resembling a circuit board or data network.



Key questions being asked about the European Union Data Strategy

1. What is the European Data Strategy and why is it important?

The European Data Strategy is the EU's comprehensive plan to create a single market for data where it can be accessed and shared securely, fairly, and efficiently. It aims to unlock the potential of data to foster innovation, support economic growth, and enhance public services. With increasing global competition and vast amounts of data generated every day, the strategy is key to ensuring Europe remains a technological leader. By enabling trusted data sharing and protecting fundamental rights, it balances economic opportunity with ethical responsibility.

2. How does the Data Strategy empower individuals and protect privacy?

The strategy ensures individuals maintain control over their data, aligning with the General Data Protection Regulation (GDPR). It supports technical tools and legal frameworks that facilitate easy data portability, allowing people to decide who accesses their data and for what purpose. Personal data spaces are envisioned to give users more oversight and control, especially over IoT-generated information. By strengthening rights and transparency, the EU reinforces trust in digital services. The strategy also limits the use of sensitive data, ensuring data is only reused when appropriate safeguards are in place, thus upholding privacy while enabling innovation.

3. What role does the strategy play in supporting businesses?

The strategy helps businesses—both large and small—gain fair and secure access to high-value data, stimulating innovation and competitiveness. It proposes clear rules and standards for data sharing to ensure legal certainty and trust among economic actors. By encouraging interoperability and common technical infrastructures, the strategy reduces fragmentation and levels the playing field, allowing startups and SMEs to compete more effectively with big tech. It also promotes voluntary data sharing, incentivizing companies to share non-personal data while safeguarding proprietary interests. The strategy ultimately supports data-driven innovation in sectors such as healthcare, manufacturing, and mobility.

4. How will the strategy benefit the public sector and citizens?

The strategy promotes better public services through enhanced access to data. Public administrations will be able to harness data insights to improve transportation, healthcare, education, and environmental management. For example, open access to real-time traffic and environmental data can lead to greener cities, while data-sharing in healthcare can support personalized treatment. Citizens stand to benefit from more responsive and efficient services, increased transparency, and broader digital participation. Furthermore, business-to-government data sharing could improve emergency responses or policymaking by providing timely, accurate information—always within the bounds of ethical use and data protection.

5. What are European common data spaces, and why are they crucial?

European common data spaces are secure, sector-specific frameworks where stakeholders can pool and access high-quality data. These spaces are designed for key sectors such as health, energy, agriculture, and mobility, promoting collaboration across borders and industries. They enable a trusted environment for sharing both public and private data while respecting privacy, trade secrets, and intellectual property. By facilitating better data availability and interoperability, common data spaces drive innovation and economic development. They also support Europe's ambitions for digital sovereignty, ensuring that critical data remains governed by EU values and is not monopolized by foreign platforms.

6. How does the Data Strategy support the European Green Deal?

The strategy supports the Green Deal by leveraging digital technologies to make Europe more sustainable. Data helps optimize energy use, manage natural resources more efficiently, and enable eco-friendly innovation. For example, precision farming using AI and environmental data reduces pesticide use, while smart grids ensure cleaner energy distribution. Additionally, data-driven insights can help track emissions, predict natural disasters, and design resilient urban infrastructure. The ICT sector itself is encouraged to reduce its carbon footprint through greener data centers and climate-neutral operations. Together, these efforts use data to accelerate the EU's goal of becoming climate-neutral by 2050.

7. What investments are planned to support the data economy?

The EU is making significant financial commitments through programs like Horizon Europe, the Digital Europe Programme (DEP), and the Connecting Europe Facility. For instance, €15 billion is earmarked for the "Digital, Industry and Space" cluster under Horizon Europe, with data and AI as central themes. DEP proposes nearly €2.5 billion for deploying data platforms and applications, including €2 billion for a High Impact project on European data spaces. These investments aim to build the infrastructure, technology, and skills needed to create a competitive and inclusive data economy while strengthening Europe's position globally.

8. How does the strategy relate to data protection laws like GDPR?

The European Data Strategy builds on the foundations of the GDPR by reinforcing individuals' control over personal data. It does not replace or weaken existing privacy rules; rather, it complements them by ensuring data is shared and reused in a lawful, ethical, and transparent manner. The strategy emphasizes non-personal and anonymized data sharing where possible, ensuring that innovation doesn't come at the expense of fundamental rights. It also promotes tools and platforms that enable safe and compliant data handling, helping citizens, businesses, and public bodies confidently navigate the evolving data economy.

9. What is the Data Act and how will it affect data sharing?

The forthcoming Data Act is a key legislative proposal under the strategy, designed to regulate fair access and use of data across the EU. It will clarify who can access what data, under what conditions, and for what purposes—particularly focusing on business-to-business (B2B) and business-to-government (B2G) scenarios. The Act aims to prevent data monopolies, encourage fair value distribution, and empower users, especially in sectors where IoT devices generate significant data. By providing legal certainty and standardized practices, the Data Act will help unlock the value of data while maintaining user trust and regulatory compliance.

10. How will the strategy foster innovation and ensure fair competition?

By breaking down data silos and encouraging interoperability, the strategy opens the door for new market entrants and innovators. It provides the legal and technical infrastructure needed for startups and SMEs to access and build on existing data, promoting a more dynamic and competitive digital economy. The strategy also reduces reliance on a handful of global tech giants by promoting open standards and EU-wide governance frameworks. Through initiatives like common data spaces, public-private partnerships, and targeted funding, it creates a fair environment where innovation can thrive while safeguarding ethical and societal values.



Understanding the European Union Data Strategy

1. Introduction



What are the core aspects

The strategy introduces Europe's vision to lead the digital transformation through a values-driven approach, with data as the key enabler of innovation and societal benefit. It stresses data sovereignty, individual empowerment, and fair economic opportunities. Central to this vision is trust in how data is collected, accessed, and used.

Why is it applicable to AI stakeholders

AI relies heavily on data availability and trust. This section's emphasis on ethical data use, transparency, and individual rights is directly relevant to AI developers, service providers, and users aiming to create responsible AI systems that enjoy public acceptance.

How does it apply to EU AI Act

The EU AI Act prioritizes trustworthy AI. The data strategy's focus on privacy, safety, and values establishes foundational conditions that AI systems must meet under the Act, including risk classification, transparency obligations, and accountability for data used in training and deployment.

2. Significance



What are the core aspects

This section outlines the data explosion, shifts in processing technologies (like edge computing), and the need for the EU to act now to secure leadership. It highlights opportunities in healthcare, sustainability, and industrial productivity, and warns against non-EU data monopolies and missed innovation chances.

Why is it applicable to AI stakeholders

For AI stakeholders, timely access to relevant, high-quality data is essential. This section underscores the urgency of creating local, competitive ecosystems where AI solutions can flourish independently from foreign dominance and within clear ethical guidelines.

How does it apply to EU AI Act

The AI Act complements this urgency by regulating high-risk AI use cases and supporting innovation in areas such as healthcare and public service. Both push for European autonomy in AI systems' development, deployment, and oversight, ensuring alignment with EU democratic and ethical values.

3. Vision



What are the core aspects

This vision describes a seamless European single market for data, where personal and non-personal data can be securely accessed, shared, and reused. It emphasizes interoperability, privacy, and open access governed by shared European values and rules.

Why is it applicable to AI stakeholders

AI systems thrive on data variety and scale. A unified data space offers legal clarity, rich datasets, and easier cross-sector access—all critical for training, testing, and improving AI models while reducing compliance complexity for developers and users.

How does it apply to EU AI Act

The AI Act relies on trustworthy data sources to determine an AI system's risk level and regulatory obligations. A common data space supports compliance by ensuring transparent, high-quality data pools are available to all AI developers, especially in regulated, high-risk domains.

4. Challenges



What are the core aspects

Key issues identified include fragmented regulations, limited data sharing between sectors, lack of incentives, unclear data ownership, and dominance by Big Tech. These hinder innovation, restrict SMEs, and reduce the EU's ability to scale data-driven services effectively and securely.

Why is it applicable to AI stakeholders

AI creators need data interoperability, fairness, and legal certainty. This section's focus on data fragmentation and imbalances speaks to the challenges faced by AI firms navigating multi-jurisdictional regulations and closed data environments, which limit fair access and model development.

How does it apply to EU AI Act

The AI Act reinforces efforts to resolve data fragmentation, particularly by demanding documentation and transparency for high-risk systems. Addressing structural data challenges aligns directly with compliance pathways under the Act, particularly for startups and SMEs using co-generated or sector-specific data.

5. Strategic Pillars



What are the core aspects

The strategy is structured around four pillars: cross-sector governance, infrastructure investment, individual empowerment, and sectoral data spaces. These pillars aim to create an inclusive, interoperable, and competitive data ecosystem that serves businesses, governments, and individuals.

Why is it applicable to AI stakeholders

These pillars give AI stakeholders a roadmap for where and how to participate—be it through ethical data governance, joining common infrastructures, or aligning with sectoral best practices. They support development and deployment of AI in high-impact areas like health, finance, and mobility.

How does it apply to EU AI Act

Each pillar complements the AI Act's objectives. For instance, governance structures support conformity assessments; infrastructure investments help meet performance and security standards; and sectoral spaces provide the high-quality training data required for high-risk AI use cases as outlined by the Act.

6. Cross-Sectoral Governance (Pillar A)



What are the core aspects

This pillar proposes a legislative framework to govern data access, sharing rights, re-use, and interoperability. It also outlines the potential Data Act and governance for common European data spaces, ensuring fair and responsible use across sectors and Member States.

Why is it applicable to AI stakeholders

AI developers need legal certainty about data use—especially for training models. This governance framework clarifies rights around co-generated data and incentivizes voluntary sharing, making it easier for suppliers, users, and third parties to build trustworthy, collaborative AI solutions.

How does it apply to EU AI Act

The AI Act requires high-risk AI systems to use datasets that are complete, accurate, and representative. The proposed governance frameworks and legal clarity from this pillar directly support those requirements, aiding conformity, risk management, and fairness under the Act's rules.

7. Data Infrastructure and Investments (Pillar B)



What are the core aspects

This section focuses on EU investment in cloud services, edge computing, federated infrastructures, and secure data-sharing platforms. These investments are essential for Europe's technological sovereignty and the capacity to host high-performance, scalable, and secure data services.

Why is it applicable to AI stakeholders

Reliable infrastructure is essential for AI model training and deployment. Investments in EU-based cloud and data-sharing systems reduce dependency on non-EU providers and offer AI businesses local, secure, and standards-compliant environments to scale AI-driven services and innovations.

How does it apply to EU AI Act

"High-risk AI systems require auditability, traceability, and security in data processing. This infrastructure roadmap supports those requirements, particularly through energy-efficient, secure, and transparent platforms that help AI providers comply with the AI Act's technical and organizational safeguards.

"

8. Empowering People and SMEs (Pillar C)



What are the core aspects

This pillar emphasizes data portability, user control tools, skills training, and support for SMEs. It promotes digital inclusion and aims to increase general and specialist data literacy while supporting legal clarity and access to services for small-scale innovators.

Why is it applicable to AI stakeholders

AI suppliers can benefit from SME-focused initiatives that improve access to data and skills. For AI users, especially consumers, enhanced control over personal data fosters trust. AI makers must design user-centered systems that respect these new rights and expectations.

How does it apply to EU AI Act

The AI Act requires transparency, especially in high-risk or user-facing systems. Empowering individuals aligns with transparency requirements such as clear disclosures, consent mechanisms, and opt-outs. For SMEs, this support enhances their ability to meet the Act's technical and legal standards.

9. Sectoral Data Spaces (Pillar D)



What are the core aspects

The EU will establish common data spaces in strategic sectors like health, mobility, energy, and public services. These spaces will pool high-quality data, develop shared governance, and support tailored infrastructure and policy frameworks to foster innovation in each domain.

Why is it applicable to AI stakeholders

Sectoral AI applications—such as diagnostic tools, smart grids, or intelligent transport—require domain-specific datasets. These curated spaces provide AI developers and suppliers with reliable, interoperable data environments that reduce barriers to entry and promote responsible development.

How does it apply to EU AI Act

Many of these sectors fall under the AI Act's "high-risk" classification. Access to regulated, high-quality datasets within these spaces helps AI providers meet the Act's obligations around accuracy, bias mitigation, and traceability, while enabling faster deployment in critical EU policy areas.

10. International Data Strategy



What are the core aspects

This section outlines the EU's stance on cross-border data flows: open, but grounded in its privacy and security standards. It promotes global alignment with EU values, while monitoring and challenging unjustified restrictions imposed by other jurisdictions.

Why is it applicable to AI stakeholders

AI developers and suppliers working across borders must manage compliance risks. This section helps stakeholders navigate international obligations, plan lawful data transfers, and align with evolving global norms. It also opens doors for compliant foreign firms to enter the EU market.

How does it apply to EU AI Act

The AI Act applies to all providers offering AI in the EU, even if based abroad. This section supports those obligations by asserting EU jurisdictional reach, demanding compliance from third-country providers, and encouraging international adoption of aligned ethical AI standards.

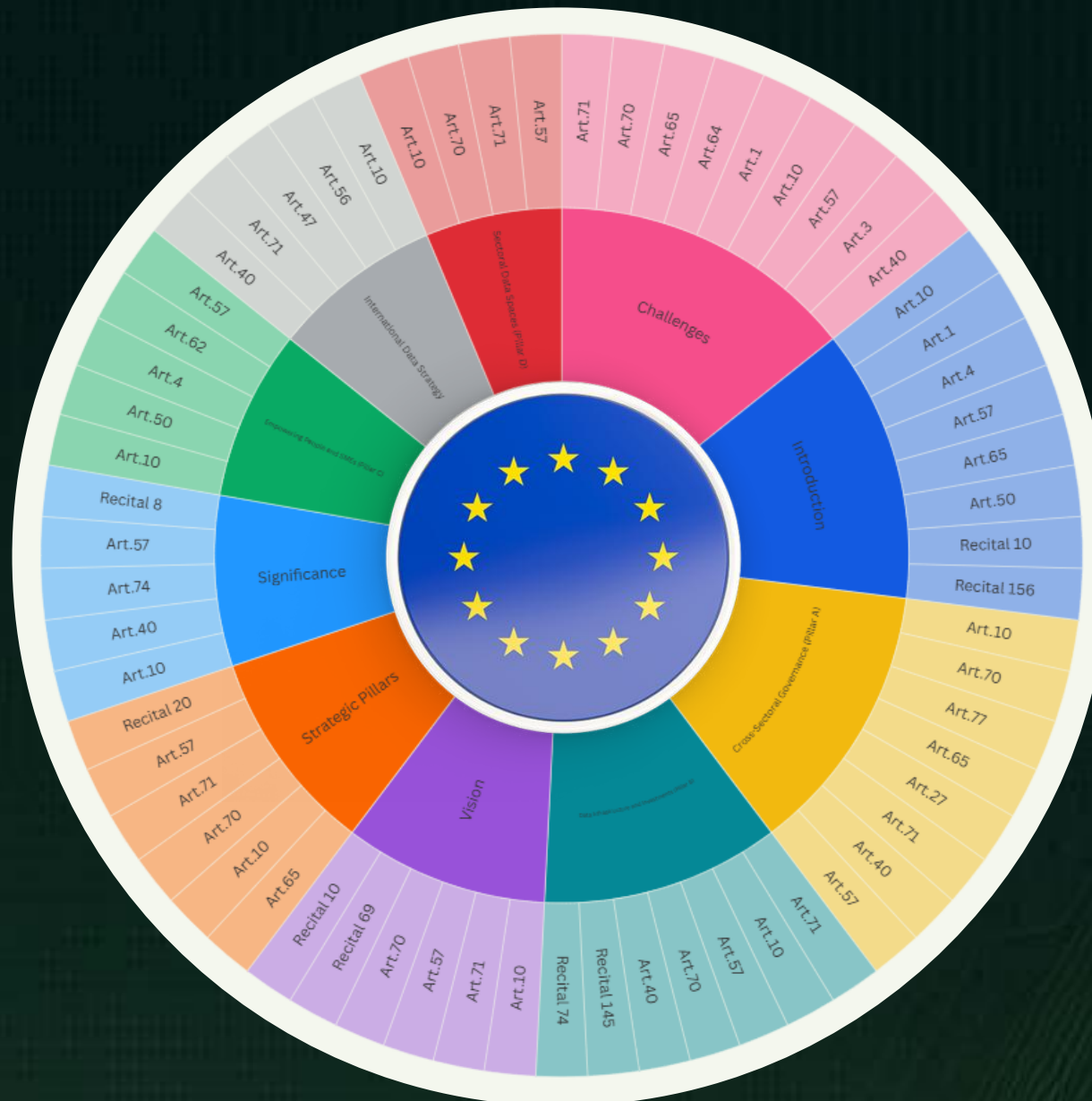


Relationship Between European Union Data Strategy and the EU AI Act

European Union Data Strategy

Vs.

EU AI Act



EU Data Strategy		EU AI Act	
Section	Description	Article(s)	Explanation
Introduction	The EU recognizes data as a critical driver of innovation, economic growth, and societal progress. This section outlines the need for a European data strategy that balances opportunity with responsibility, emphasizing individual rights, data-driven innovation, and Europe's potential to lead a human-centric, trusted digital economy.	10, 1, 4, 57, 65, 50, Recital 10, Recital 156	The European Union (EU) acknowledges the pivotal role of data in fostering innovation, economic growth, and societal advancement. The EU AI Act operationalizes elements of the European Data Strategy by establishing a framework that balances opportunities with responsibilities, emphasizing individual rights, data-driven innovation, and Europe's potential to lead a human-centric, trusted digital economy. Key Provisions of the EU AI Act 1. Data Governance and Quality - High-Quality Data Sets: The Act mandates that high-risk AI systems be developed using high-quality data sets for training, validation, and testing. These data sets must be relevant, representative, and free of errors to ensure the AI system performs safely and as intended, without discrimination. - Data Governance Practices: Providers must implement data governance and management practices that address data collection, preparation, and bias detection to protect fundamental rights and prevent discrimination . 2. Protection of Individual Rights - Privacy and Data Protection: The Act aligns with existing EU data protection laws, such as the General Data Protection Regulation (GDPR), ensuring that personal data is processed lawfully and that individual rights are safeguarded throughout the AI system's lifecycle - Bias Detection and Correction: Providers are required to detect and mitigate biases in AI systems to prevent negative impacts on fundamental rights and discrimination. 3. Innovation and Economic Growth - AI Regulatory Sandboxes: The Act introduces AI regulatory sandboxes to foster innovation by allowing AI systems to be developed and tested in a controlled environment. This supports the development of innovative AI solutions while ensuring compliance with regulatory requirements. - Support for SMEs and Start-ups: The Act includes measures to support innovation, particularly for small and medium-sized enterprises (SMEs) and start-ups, by providing guidance and facilitating access to resources and expertise

			4. Human-Centric and Trusted AI - Transparency Obligations: Providers and deployers of certain AI systems must ensure transparency by informing users when they are interacting with an AI system and disclosing AI-generated content. - AI Literacy: The Act emphasizes the importance of AI literacy to equip individuals with the knowledge to make informed decisions about AI systems, promoting a human-centric approach. 5. Governance and Compliance - European Artificial Intelligence Board: The Act establishes a governance framework, including the European Artificial Intelligence Board, to coordinate and support the application of the regulation at both national and Union levels - Market Surveillance and Compliance: The Act ensures that market surveillance authorities have the necessary powers to enforce compliance, thereby maintaining trust in AI systems.
Significance	With data volumes exploding and technologies evolving rapidly, the EU must seize this moment to become a global leader. The section emphasizes the importance of managing this transformation to ensure competitiveness, sustainability, and ethical data use, while avoiding dominance by non-European players.	10, 40, 74, 57, Recital 8	The European Union (EU) aims to harness the burgeoning data landscape to establish itself as a global leader in AI, ensuring competitiveness, sustainability, and ethical data use. The EU AI Act operationalizes these goals by setting a comprehensive framework that addresses data governance, innovation, and ethical considerations. Key Provisions of the EU AI Act 1. Data Governance and Quality - High-Quality Data Sets: The Act mandates that high-risk AI systems use high-quality data sets for training, validation, and testing. These data sets must be relevant, representative, and free of errors to ensure the AI system performs safely and without discrimination - Data Governance Practices: Providers must implement robust data governance practices, including bias detection and correction, to protect fundamental rights and prevent discrimination. 2. Innovation and Competitiveness - AI Regulatory Sandboxes: The Act introduces AI regulatory sandboxes to foster innovation by allowing AI systems to be developed and tested in a controlled environment. This supports the development of innovative AI solutions while ensuring compliance with regulatory requirements. - Support for SMEs and Start-ups: The Act includes measures to support innovation, particularly for small and medium-sized enterprises (SMEs) and start-ups, by providing guidance and facilitating access to resources and expertise.

			3. Ethical and Sustainable AI - Transparency and Accountability: The Act requires transparency in AI systems, ensuring users are informed when interacting with AI and that AI-generated content is disclosed - Ethical Guidelines: The Act aligns with the EU's ethical guidelines for trustworthy AI, promoting a human-centric approach that respects fundamental rights and values. 4. Avoiding Dominance by Non-European Players - Harmonized Standards: The Act establishes harmonized standards to ensure a level playing field for European AI developers, preventing dominance by non-European players and fostering a competitive internal market. - Market Surveillance and Compliance: The Act ensures that market surveillance authorities have the necessary powers to enforce compliance, maintaining trust in AI systems and preventing market dominance by non-compliant entities.
Vision	The strategy envisions a unified European data space where data flows freely across sectors and borders under fair rules. It aims to empower people, businesses, and governments to use data responsibly and effectively, while protecting privacy and boosting innovation across the economy.	10, 71, 57, 70, Recital 69, Recital 10	Introduction The European Data Strategy envisions a unified data space where data flows freely across sectors and borders, empowering stakeholders while ensuring privacy and innovation. The EU AI Act operationalizes this vision by establishing a framework that promotes responsible data use, protects privacy, and fosters innovation. Key Provisions of the EU AI Act 1. Data Governance and Free Flow - High-Quality Data Sets: The Act mandates that high-risk AI systems use high-quality data sets for training, validation, and testing. These data sets must be relevant, representative, and free of errors, ensuring the AI system performs safely and without discrimination. - EU Database for High-Risk AI Systems: The Act establishes an EU database to register high-risk AI systems, facilitating transparency and accountability. This database ensures that information about AI systems is accessible and can flow freely across borders. 2. Privacy and Data Protection - Alignment with GDPR: The Act aligns with existing EU data protection laws, such as the General Data Protection Regulation (GDPR), ensuring that personal data is processed lawfully and individual rights are

			safeguarded. - Data Minimization and Protection by Design: The Act emphasizes principles of data minimization and protection by design, ensuring that personal data is processed responsibly throughout the AI system's lifecycle. 3. Innovation and Economic Growth - AI Regulatory Sandboxes: The Act introduces AI regulatory sandboxes to foster innovation by allowing AI systems to be developed and tested in a controlled environment. This supports the development of innovative AI solutions while ensuring compliance with regulatory requirements. - Support for SMEs and Start-ups: The Act includes measures to support innovation, particularly for small and medium-sized enterprises (SMEs) and start-ups, by providing guidance and facilitating access to resources and expertise. 4. Cross-Sector and Cross-Border Data Flow - Harmonized Standards: The Act establishes harmonized standards to ensure a level playing field for European AI developers, facilitating the free flow of data across sectors and borders. - Cooperation and Coordination: The Act promotes cooperation and coordination among Member States and relevant authorities to ensure consistent application and enforcement of AI regulations across the EU.
Challenges	Key barriers include fragmented national approaches, limited data sharing (especially B2G and B2B), dominance of Big Tech, and insufficient interoperability. The strategy also notes the need for better infrastructure, legal clarity, and governance to realize the full potential of the EU's data economy.	71, 70, 65, 64, 1, 10, 57, 3, 40	Introduction The European Data Strategy identifies several challenges, including fragmented national approaches, limited data sharing, dominance of Big Tech, and insufficient interoperability. The EU AI Act addresses these challenges by establishing a unified framework to enhance data sharing, ensure legal clarity, and improve governance. Key Provisions of the EU AI Act 1. Addressing Fragmentation - Harmonized Rules: The Act provides harmonized rules for the development, market placement, and use of AI systems across the EU, reducing fragmentation and ensuring a consistent approach. - European Artificial Intelligence Board: The establishment of the Board facilitates coordination among Member States, promoting a unified approach to AI governance.

			2. Enhancing Data Sharing - EU Database for High-Risk AI Systems: The Act mandates the creation of an EU database to register high-risk AI systems, promoting transparency and facilitating data sharing across borders. - Data Governance Requirements: High-risk AI systems must adhere to stringent data governance practices, ensuring data quality and facilitating interoperability. 3. Mitigating Dominance of Big Tech - Market Surveillance and Compliance: The Act empowers national authorities to monitor AI systems, ensuring compliance and preventing market dominance by non-compliant entities. - Support for SMEs and Start-ups: The Act includes measures to support innovation, particularly for SMEs and start-ups, leveling the playing field and reducing the dominance of large tech companies. 4. Improving Interoperability and Infrastructure - AI Regulatory Sandboxes: These sandboxes provide a controlled environment for testing AI systems, fostering innovation and improving interoperability by allowing cross-border cooperation. - Harmonized Standards: The Act promotes the use of harmonized standards to ensure interoperability and facilitate the integration of AI systems across different sectors. 5. Legal Clarity and Governance - Clear Definitions and Obligations: The Act provides clear definitions and obligations for AI system providers and deployers, ensuring legal clarity and reducing uncertainty. - Governance Framework: The Act establishes a comprehensive governance framework, including national competent authorities and the AI Office, to oversee the implementation and enforcement of AI regulations.
Strategic Pillars	The data strategy is structured around four pillars: (A) a cross-sectoral governance framework, (B) infrastructure and technology investments, (C) individual empowerment and skills, and (D) sector-specific European data spaces. These coordinated efforts	65, 10, 70, 71, 57, Recital 20	Introduction The European Data Strategy is built on four strategic pillars: a cross-sectoral governance framework, infrastructure and technology investments, individual empowerment and skills, and sector-specific European data spaces. The EU AI Act operationalizes these pillars by establishing a comprehensive framework that promotes governance, supports infrastructure, empowers individuals, and facilitates sector-specific data spaces.

	aim to create a thriving, competitive, and inclusive data-driven economy.		Key Provisions of the EU AI Act 1. Cross-Sectoral Governance Framework - European Artificial Intelligence Board: The Act establishes the European Artificial Intelligence Board to coordinate and support the application of AI regulations across Member States, ensuring a unified approach to governance. - National Competent Authorities: Each Member State is required to designate national competent authorities to oversee the implementation and enforcement of the AI Act, ensuring consistency and coordination at the national level. 2. Infrastructure and Technology Investments - AI Regulatory Sandboxes: The Act mandates the establishment of AI regulatory sandboxes to foster innovation and facilitate the development and testing of AI systems in a controlled environment. These sandboxes support infrastructure development and technological advancements. - EU Database for High-Risk AI Systems: The creation of an EU database for high-risk AI systems enhances infrastructure by providing a centralized platform for information sharing and transparency. 3. Individual Empowerment and Skills - AI Literacy: The Act emphasizes the importance of AI literacy, equipping individuals with the knowledge to make informed decisions about AI systems. This empowerment is crucial for fostering a data-driven economy. - Transparency and Accountability: The Act requires transparency in AI systems, ensuring users are informed when interacting with AI and that AI-generated content is disclosed, thereby empowering individuals. 4. Sector-Specific European Data Spaces - Data Governance and Quality: The Act mandates high-quality data governance practices for high-risk AI systems, ensuring data sets are relevant, representative, and free of errors. This supports the creation of sector-specific data spaces by ensuring data quality and interoperability. - Harmonized Standards: The Act promotes the use of harmonized standards to facilitate interoperability and integration of AI systems across different sectors, supporting sector-specific data spaces.
Cross-Sectoral	This pillar proposes legislative and regulatory frameworks for data	10, 70, 77, 65, 27, 71, 40, 57	Introduction

Governance (Pillar A)	access, use, and sharing. It includes initiatives like the Data Act and high-value dataset mandates. Governance will support interoperability, fair data use, and mechanisms for responsible data sharing between entities and across borders.	The European Data Strategy's Cross-Sectoral Governance pillar emphasizes legislative and regulatory frameworks for data access, use, and sharing. The EU AI Act operationalizes this pillar by establishing governance structures and regulations that promote interoperability, fair data use, and responsible data sharing across borders. Key Provisions of the EU AI Act 1. Governance Framework - European Artificial Intelligence Board: The Act establishes the European Artificial Intelligence Board to coordinate and support the application of AI regulations across Member States, ensuring a unified approach to governance. - National Competent Authorities: Each Member State is required to designate national competent authorities to oversee the implementation and enforcement of the AI Act, ensuring consistency and coordination at the national level. 2. Data Access and Sharing - EU Database for High-Risk AI Systems: The Act mandates the creation of an EU database to register high-risk AI systems, promoting transparency and facilitating data sharing across borders. - Data Governance Requirements: High-risk AI systems must adhere to stringent data governance practices, ensuring data quality and facilitating interoperability. 3. Interoperability and Fair Data Use - Harmonized Standards: The Act promotes the use of harmonized standards to ensure interoperability and facilitate the integration of AI systems across different sectors. - AI Regulatory Sandboxes: These sandboxes provide a controlled environment for testing AI systems, fostering innovation and improving interoperability by allowing cross-border cooperation. 4. Responsible Data Sharing - Transparency and Accountability: The Act requires transparency in AI systems, ensuring users are informed when interacting with AI and that AI-generated content is disclosed, thereby promoting responsible data sharing. - Market Surveillance and Compliance: The Act empowers national authorities to monitor AI systems, ensuring compliance and preventing market dominance by non-compliant entities.
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Data Infrastructure and Investments (Pillar B)	The EU will invest in shared, federated, and secure data infrastructures, including high-impact projects on European data spaces and cloud services. Emphasis is placed on developing trustworthy technologies and standards that serve public, private, and research needs while ensuring energy efficiency and security.	71, 10, 57, 70, 40, Recital 145, Recital 74	Introduction The European Data Strategy's Data Infrastructure and Investments pillar focuses on developing shared, federated, and secure data infrastructures. The EU AI Act operationalizes this pillar by promoting investments in data infrastructure, fostering innovation, and ensuring the development of trustworthy technologies and standards. Key Provisions of the EU AI Act 1. Investment in Data Infrastructure - AI Regulatory Sandboxes: The Act mandates the establishment of AI regulatory sandboxes to foster innovation and facilitate the development and testing of AI systems in a controlled environment. These sandboxes support infrastructure development by providing a space for experimentation and collaboration. - EU Database for High-Risk AI Systems: The creation of an EU database for high-risk AI systems enhances infrastructure by providing a centralized platform for information sharing and transparency. 2. Development of Trustworthy Technologies - Harmonized Standards: The Act promotes the use of harmonized standards to ensure interoperability and facilitate the integration of AI systems across different sectors. This supports the development of trustworthy technologies that meet high standards of safety and performance - Data Governance and Quality: High-risk AI systems must adhere to stringent data governance practices, ensuring data quality and facilitating interoperability. This supports the development of reliable and secure AI technologies. 3. Energy Efficiency and Security - Energy-Efficient AI Models: The Act encourages the development of energy-efficient AI models, promoting sustainability and reducing the environmental impact of AI systems. - Cybersecurity Measures: The Act requires high-risk AI systems to meet appropriate levels of cybersecurity, ensuring the protection of data and the integrity of AI systems 4. Support for Public, Private, and Research Needs - AI-on-Demand Platform and Innovation Hubs: The Act supports initiatives like the AI-on-demand
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			platform and European Digital Innovation Hubs, which provide technical and scientific support to AI developers, fostering collaboration between public, private, and research sectors. - Collaboration and Coordination: The Act promotes cooperation among Member States and relevant authorities to ensure consistent application and enforcement of AI regulations across the EU, supporting a unified approach to data infrastructure development.
Empowering People and SMEs (Pillar C)	Individuals will be given more control over their data through tools like personal data spaces and enhanced portability. Investment in digital skills and data literacy, especially for SMEs and the general workforce, is critical to ensuring inclusive participation in the data economy.	10, 50, 4, 62, 57	Introduction The European Data Strategy's Empowering People and SMEs pillar focuses on giving individuals more control over their data and enhancing digital skills and data literacy, particularly for small and medium-sized enterprises (SMEs) and the general workforce. The EU AI Act operationalizes this pillar by establishing frameworks and initiatives that enhance data control, promote digital skills, and support SMEs. Key Provisions of the EU AI Act 1. Data Control and Portability - Transparency Obligations: The Act mandates transparency for AI systems, ensuring that individuals are informed when interacting with AI and that AI-generated content is disclosed. This transparency empowers individuals by providing them with more control over their data interactions. - Data Governance and Quality: High-risk AI systems must adhere to stringent data governance practices, ensuring data quality and facilitating interoperability, which supports data portability and control. 2. Investment in Digital Skills and Data Literacy - AI Literacy: The Act emphasizes the importance of AI literacy, equipping individuals with the knowledge to make informed decisions about AI systems. This empowerment is crucial for fostering inclusive participation in the data economy. - Support for SMEs: The Act includes measures to support SMEs, including priority access to AI regulatory sandboxes and tailored training activities, enhancing their digital skills and capacity to engage in the data economy. 3. Support for SMEs and Workforce - AI Regulatory Sandboxes: These sandboxes provide a controlled environment for testing AI systems, fostering innovation and supporting SMEs by offering guidance and resources to develop AI solutions 30. - Conformity Assessment and Fees: The Act takes into account the specific interests and needs of SMEs

			when setting conformity assessment fees, reducing financial barriers and supporting their participation in the AI ecosystem.
Sectoral Data Spaces (Pillar D)	Nine sectoral data spaces will be developed in areas such as health, mobility, energy, and public administration. Each space will address domain-specific challenges and enable targeted data use, with the goal of enhancing innovation, service delivery, and cross-sector interoperability.	10, 70, 71, 57	Introduction The European Data Strategy's Sectoral Data Spaces pillar focuses on developing nine sectoral data spaces in areas like health, mobility, energy, and public administration. These spaces aim to address domain-specific challenges and enhance innovation, service delivery, and cross-sector interoperability. The EU AI Act operationalizes this pillar by establishing frameworks and initiatives that support the creation and functioning of these data spaces. Key Provisions of the EU AI Act 1. Data Governance and Quality - High-Quality Data Sets: The Act mandates that high-risk AI systems use high-quality data sets for training, validation, and testing. These data sets must be relevant, representative, and free of errors, ensuring the AI system performs safely and without discrimination 1. - Data Governance Practices: Providers must implement robust data governance practices, ensuring data quality and facilitating interoperability, which is crucial for sectoral data spaces 1. 2. Support for Sectoral Data Spaces - EU Database for High-Risk AI Systems: The creation of an EU database for high-risk AI systems enhances transparency and facilitates data sharing across sectors, supporting the development of sectoral data spaces 4. - Harmonized Standards: The Act promotes the use of harmonized standards to ensure interoperability and facilitate the integration of AI systems across different sectors, which is essential for sectoral data spaces 4. 3. Innovation and Cross-Sector Interoperability - AI Regulatory Sandboxes: These sandboxes provide a controlled environment for testing AI systems, fostering innovation and supporting cross-sector interoperability by allowing collaboration and experimentation. - Collaboration and Coordination: The Act promotes cooperation among Member States and relevant authorities to ensure consistent application and enforcement of AI regulations across the EU, supporting a unified approach to sectoral data spaces 2.

			4. Addressing Domain-Specific Challenges - Sector-Specific Requirements: The Act includes provisions for addressing domain-specific challenges by ensuring that AI systems meet high standards of safety and performance, tailored to the needs of each sector 1.
International Data Strategy	The EU supports free but safe international data flows based on European values and standards. It will combat unjust restrictions abroad while promoting global adoption of its ethical data governance model. A data flow measurement framework will also guide policy and investment.	40, 71, 47, 56, 10	Introduction The European Data Strategy's International Data Strategy emphasizes free yet secure international data flows, grounded in European values and standards. The EU AI Act operationalizes this strategy by establishing frameworks that promote ethical data governance, ensure compliance with European standards, and facilitate international cooperation. Key Provisions of the EU AI Act 1. Promoting Ethical Data Governance - Harmonized Standards: The Act mandates the use of harmonized standards for high-risk AI systems, ensuring that AI systems conform to European values and standards. This promotes ethical data governance and facilitates international adoption of these standards. - Codes of Practice: The AI Office encourages the development of codes of practice at the Union level, which contribute to the proper application of the regulation and align with international approaches. 2. Ensuring Compliance with European Standards - EU Declaration of Conformity: Providers of high-risk AI systems must draw up an EU declaration of conformity, stating that the system meets the requirements set out in the regulation. This ensures that AI systems comply with European standards and facilitates their acceptance internationally. - Data Governance Requirements: High-risk AI systems must adhere to stringent data governance practices, ensuring data quality and compliance with European data protection laws. 3. Facilitating International Cooperation - EU Database for High-Risk AI Systems: The establishment of an EU database for high-risk AI systems enhances transparency and facilitates data sharing across borders, supporting international cooperation. - International Collaboration: The Act supports international collaboration by aligning with existing international standards and promoting global cooperation on standardization.

			4. Combating Unjust Restrictions Abroad - Market Surveillance and Compliance: The Act empowers national authorities to monitor AI systems, ensuring compliance and preventing unjust restrictions on data flows. - Global Adoption of Standards: By promoting harmonized standards and ethical data governance, the Act encourages global adoption of European standards, combating unjust restrictions abroad.
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Calls to Action



Develop a Strategic Data Readiness Plan

Companies must assess their data infrastructure and capabilities to capitalize on growing volumes and edge computing. This involves evaluating current storage, processing, and analytics tools, forecasting future needs, and aligning with industry shifts toward decentralized and real-time data-driven services and products



Join or Build Sectoral Data Alliances

Collaborate with peers, academia, or public bodies to contribute to and benefit from the European data space. Enterprises can pool non-sensitive data, co-develop industry standards, and access shared datasets, fostering innovation and ensuring they operate within future regulatory and technological expectations



Audit and Expand Data Sharing Policies

Review internal and external data-sharing practices, identifying gaps that prevent optimal use. Companies should create incentives for B2B sharing, develop protocols for anonymized B2G contributions, and actively address trust, liability, and competitive concerns that limit collaborative data usage



Align with Four Pillars Strategy

Enterprises should map their digital transformation initiatives to the EU's four pillars: governance, infrastructure, skills, and sectoral engagement. This involves setting measurable goals across each pillar, from cloud adoption and data sharing policies to staff training and involvement in sector-specific data ecosystems

Conclusion

The European Union Data Strategy represents a fundamental shift in how Europe organizes, shares, and governs data in the digital age. This ambitious initiative moves toward a more interoperable, transparent, and inclusive model for data-driven innovation. By advancing investments in infrastructure, common data spaces, governance frameworks, and digital skills, the strategy aims to create a trusted, competitive, and sustainable data economy across the EU.

This report emphasizes how the European Data Strategy serves as a unifying framework for businesses, policymakers, and other stakeholders, fostering cross-sector collaboration and aligning data initiatives with Europe's principles of fairness, accountability, and individual empowerment. When implemented effectively, these efforts can accelerate Europe's leadership in the global data economy while ensuring compliance with stringent privacy and security standards.

However, realizing the full potential of the European Data Strategy requires more than strategic vision—it calls for coordinated execution, continuous investment, and strong public-private partnerships.

Stakeholders must align their plans with the strategy's core pillars, particularly in areas such as cross-sector governance, infrastructure readiness, SME empowerment, and sectoral data spaces. Smaller organizations may face challenges in meeting these goals, highlighting the need for early planning, regulatory clarity, and collaborative ecosystems.

Despite these challenges, the European Data Strategy offers a transformative opportunity for Europe to assert global leadership in ethical and value-driven digital innovation. As data continues to power advancements in sectors from healthcare and energy to mobility and public services, the strategy provides the essential foundation for building systems that are both cutting-edge and trustworthy. In conclusion, successful implementation of the European Data Strategy could establish a worldwide benchmark for responsible data governance, influencing future international frameworks and ensuring a resilient, inclusive digital economy for decades to come.

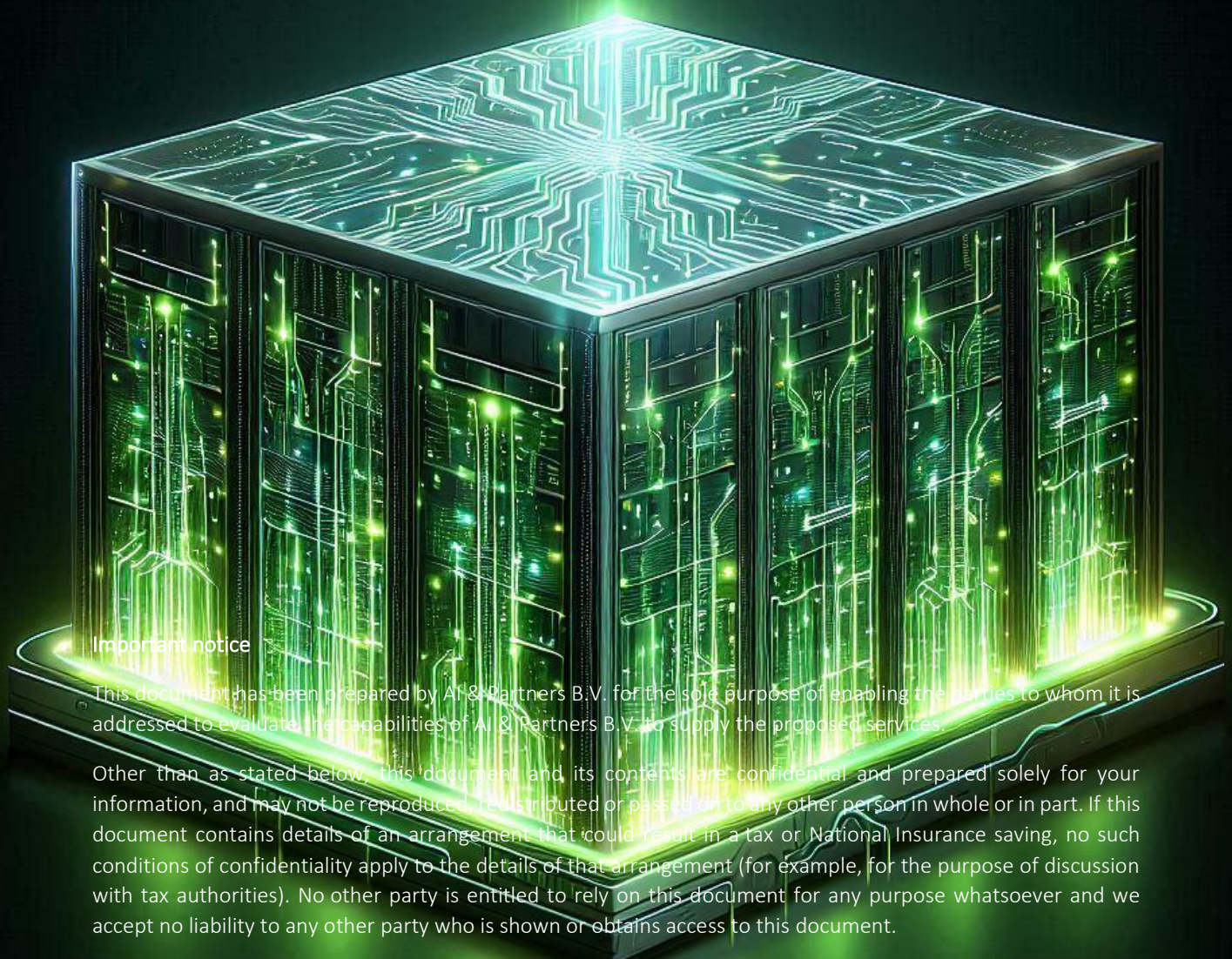
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